

Buzy AI

AI-Powered Document Intelligence for African Organizations

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GDG on Campus UCB — Build with Gemma 2026

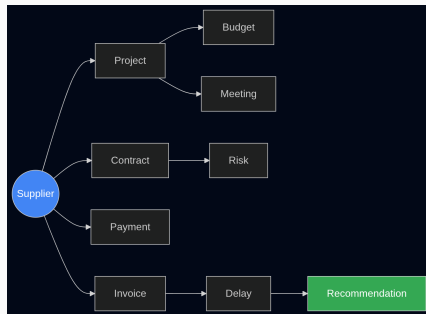
The Problem: Document Chaos in African Organizations

Across Africa, SMEs, cooperatives, and NGOs manage critical decisions using documents scattered across paper, PDFs, images, and audio recordings.

- **No centralized knowledge** — contracts, invoices, meeting notes live in silos
- **Manual analysis** takes hours or days
- **Institutional knowledge walks away** when staff leave
- **Language barriers** — French, English, Lingala, Swahili documents mixed

Real questions organizations ask:

“Which supplier is most reliable for our co-operative?”



Why This Matters for DRC & Africa

- **80% of African SMEs** operate without formal document management systems
- **Manual contract review** costs small businesses \$500+ per contract in lost productivity
- **Meeting decisions** are lost when only recorded as audio — no transcription, no search
- **Multilingual reality** — French administration, local languages for operations

Buzy AI bridges the gap: no expensive ERP, no cloud subscription — runs on a laptop, works offline, speaks French and English.



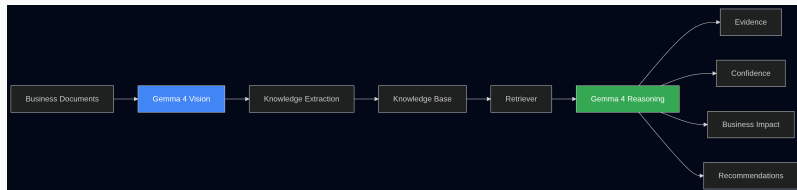
Solution: Buzy AI

Buzy AI is a multimodal document intelligence system that transforms raw business documents into structured, explainable decisions.

1. **Ingest** PDFs, DOCX, images (receipts, dashboards), and audio recordings
2. **Extract** structured facts using Gemma 4 Vision-Language
3. **Build** a searchable knowledge base
4. **Retrieve** relevant evidence for any business question
5. **Generate** structured recommendations with source citations



Architecture



Raw documents →

multimodal analysis → fact extraction → knowledge base → reasoning →
recommendations

How Gemma 4 Powers the Solution

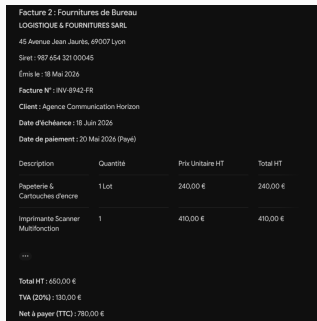
Gemma 4 Capabilities Used:

- **Vision-Language:** Read scanned invoices, receipts, handwritten notes, dashboards
- **Multilingual:** Understand French, English, and code-switched documents
- **Structured generation:** Output JSON with evidence, reasoning, confidence
- **Reasoning:** Chain-of-thought for explainable business decisions
- **LoRA fine-tuning:** Adapted for structured business reasoning

Invoice Extraction with Gemma 4 Vision

Input: Scanned invoice image

Output: { "supplier": "...",
 "amount": "42 300 USD",
 "due_date": "2026-06-01",
 "status": "paid 18 days late" }

A scanned image of a French invoice. The header contains the supplier's name 'Facture 2 : Fournitures de Bureau LOGISTIQUE & FOURNITURES SARL', address '45 Avenue Jean Jaurès, 69007 Lyon', and SIRET number '987 654 321 00045'. It is dated 'Émis le : 18 Mai 2026' and has invoice number 'Facture N° : INV-8942-FR'. The client is 'Agence Communication Horizon' and the due date is 'Date d'échéance : 18 Juin 2026'. The payment terms are 'Date de paiement : 20 Mai 2026 (Payé)'. The main body is a table with 4 columns: Description, Quantité, Prix Unitaire HT, and Total HT. It lists two items: 'Papeterie & Cartouches d'encre' (1 Lot, 240,00 €) and 'Imprimante Scanner Multifonction' (1, 410,00 €). The footer shows the totals: 'Total HT : 650,00 €', 'TVA (20%) : 130,00 €', and 'Net à payer (TTC) : 780,00 €'.

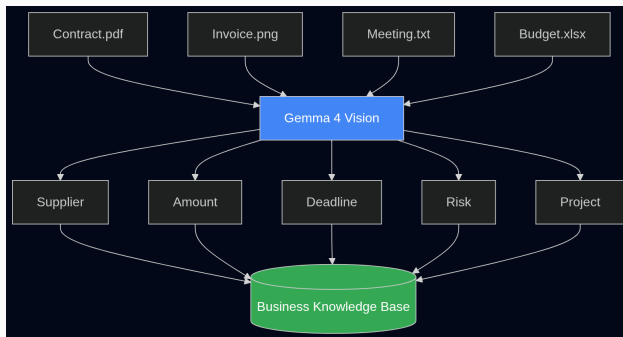
Facture 2 : Fournitures de Bureau LOGISTIQUE & FOURNITURES SARL 45 Avenue Jean Jaurès, 69007 Lyon Siret : 987 654 321 00045 Émis le : 18 Mai 2026 Facture N° : INV-8942-FR Client : Agence Communication Horizon Date d'échéance : 18 Juin 2026 Date de paiement : 20 Mai 2026 (Payé)			
Description	Quantité	Prix Unitaire HT	Total HT
Papeterie & Cartouches d'encre	1 Lot	240,00 €	240,00 €
Imprimante Scanner Multifonction	1	410,00 €	410,00 €
...			
Total HT : 650,00 € TVA (20%) : 130,00 € Net à payer (TTC) : 780,00 €			

Fine-Tuning with LoRA



- Rank 8 LoRA adapters on language layers
- Trained on 4 structured reasoning examples
- 20 epochs, final loss 0.299
- Only 0.25% of parameters trained (12.6M of 5.1B)
- Training: 57 seconds on Kaggle (2x T4)
- Weights: DestinBir/buzy-ai-gemma4-lora

Knowledge Extraction Pipeline



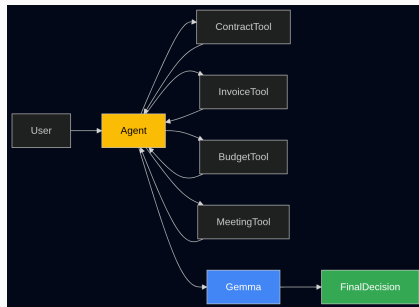
Entities extracted: suppliers,
contracts, invoices, projects, delivery risks, budget variances

Retrieval-Augmented Reasoning



1. Facts retrieved from knowledge base by keyword
2. Gemma 4 generates reasoning chain
3. Output: evidence → reasoning → confidence → impact → actions

AI Agent with Tool Use



- Gemma 4 selects which business tool to call based on user question
- Tools: `get_invoice_status()`, `get_contract_expiry()`
- Tool result fed back for final answer generation

Live Scenario: Supplier Risk Analysis

Context: A Congolese procurement cooperative manages 6 suppliers

Question: Which supplier represents the highest operational risk?

- **Evidence:** 18-day payment delay average, delivery instability flagged in 2 meetings, contract expiring in 27 days
- **Reasoning:** Repeated delays + sole-source dependency on critical project
- **Confidence:** High (0.82)

Impact: 2-week project delay risk; unfavorable renewal terms

Actions:

1. Begin renewal negotiations immediately
2. Qualify a secondary supplier
3. Escalate payment approvals

All recommendations cite exact source documents

Gemma 4 enables business analysis in both French and English:

English

"Which supplier represents the highest operational risk next quarter?"

→ Structured JSON with evidence, reasoning, confidence

French

"Quel fournisseur représente le plus grand risque opérationnel le trimestre prochain ?"

→ Same JSON structure in French

Critical for DRC: French administration + local languages in daily operations

Deployed on Hugging Face Spaces:

- Upload images, PDF/DOCX/TXT, audio
- Ask any business question in English or French
- One-click examples (invoice, contract, meeting)
- Source-attributed Markdown reports

`python app.py`
→ localhost:7860

Features:

- Lazy model loading (fast startup)
- Gradio progress tracking
- CUDA auto-detection
- Whisper ASR for meeting audio
- Works on CPU (no GPU required)

Layer	Technology
Base Model	Gemma 4 (E2B-it) 4-bit quantized
Fine-tuning	LoRA via Unsloth
Fine-tuned Weights	DestinBir/buzy-ai-gemma4-lora
Web App	Gradio on Hugging Face Spaces
Speech-to-Text	Whisper (openai/whisper-small)
Document Parsing	PyMuPDF, python-docx
Training Framework	TRL (SFTTrainer)
Training Platform	Kaggle (2x T4 GPU)

Metric	Value
Base Model	Gemma 4 (E2B-it)
Fine-tuning Method	LoRA (rank 8)
Trainable Parameters	12.6M / 5.1B (0.25%)
Training Epochs	20
Final Loss	0.299
Training Time	~57s (2x T4)
Inference	CPU or CUDA
Deployment	Hugging Face Spaces

Current impact:

- Transforms hours of manual document review into seconds
- Source-attributed output builds trust and verifiability
- Runs locally — no internet required after setup
- Supports French and English for DRC context

Future work:

- Add Lingala and Swahili support
- Mobile app for field data collection
- Voice interface for low-literacy users
- Integration with local accounting software (e.g., Sage, QuickBooks)
- Offline-first with PWA
- Fine-tune on real African business documents

github github.com/DestinBir/Buzy_Multimodal_Gemma

huggingface huggingface.co/spaces/DestinBir/buzy-ai

huggingface Model: [DestinBir/buzy-ai-gemma4-lora](#)

kaggle [Kaggle Notebook](#)

Thank you!

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Built with Gemma 4 by Google